“Movie Dataset Analysis using SQL”

📌 **Project Description**

This project focuses on analyzing a movie dataset using SQL to extract meaningful insights. The objective is to understand the structure of the data, perform data cleaning, and explore patterns such as trends in release years, missing values, and distribution of different attributes.

SQL is used to query the dataset, check data quality, and answer analytical questions that are useful for decision-making. This project demonstrates practical SQL skills required for a data analyst role.

🔍 **Step 1: Understanding Dataset Structure**

✅ **Query:**

DESCRIBE titles;

Result Grid | Filter Rows: | Export: | Wrap

Field	Type	Null	Key	Default	Extra
id	text	YES		NULL	
title	text	YES		NULL	
type	text	YES		NULL	
description	text	YES		NULL	
release_year	int	YES		NULL	
age_certification	text	YES		NULL	
runtime	int	YES		NULL	
genres	text	YES		NULL	
production_countries	text	YES		NULL	
seasons	text	YES		NULL	
imdb_id	text	YES		NULL	
imdb_score	text	YES		NULL	
imdb_votes	text	YES		NULL	
tmdb_popularity	double	YES		NULL	
tmdb_score	text	YES		NULL	

📊 **Explanation:**

This query shows the structure of the dataset including column names, data types, and whether NULL values are allowed.

👉 Insight:

We understand what type of data is available and how it is stored.

📊 **Step 2: Preview the Data**

✅ **Query:**

*SELECT * FROM titles*
LIMIT 5;

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

id	title	type	description	release_year	age_certification	runtime	genres	production_countries	seasons	imdb_id	imdb_score	imdb_votes	tmdb_popularity	tmdb_score
ts300399	Five Came Back: The Reference Films	SHOW	This collection includes 1...	1945	TV-MA	51	['documentation']	['US']	1.0				0.6	
tm84618	Taxi Driver	MOVIE	A mentally unstable Viet...	1976	R	114	['drama', 'crime']	['US']		tt0075314	8.2	808582.0	40.965	8.179
tm154986	Deliverance	MOVIE	Intent on seeing the Ca...	1972	R	109	['drama', 'action', 'thriller', 'europe...	['US']		tt0068473	7.7	107673.0	10.01	7.3
tm127384	Monty Python and the Holy Grail	MOVIE	King Arthur, accompanie...	1975	PG	91	['fantasy', 'action', 'comedy']	['GB']		tt0071853	8.2	534486.0	15.461	7.811
tm120801	The Dirty Dozen	MOVIE	12 American military pris...	1967		150	['war', 'action']	['GB', 'US']		tt0061578	7.7	72662.0	20.398	7.6

📊 **Explanation:**

This query displays the first 5 rows of the dataset.

👉 Insight:

Helps to understand how the actual data looks and identify possible issues.

📌 **Business Questions Solved**

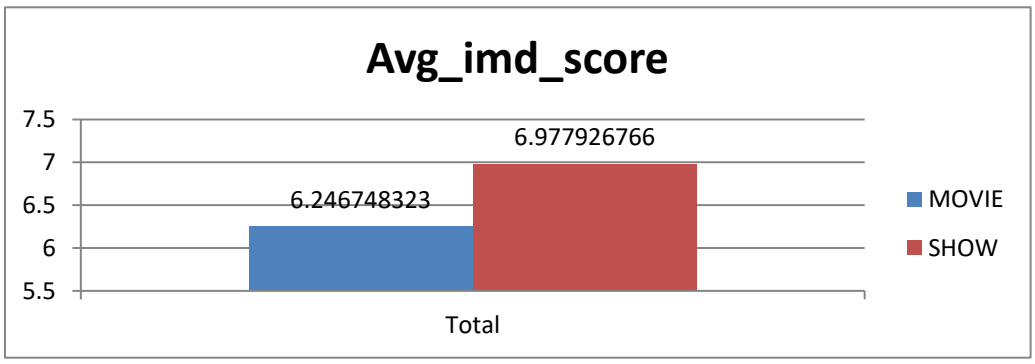
- What is the average rating of movies vs shows?
- How has content production changed over decades?
- Which age certifications have higher ratings?
- What is the average runtime of content types?
- Which shows have the most seasons?
- How many titles are released each year?

🎯 **Step 3: Average IMDB and TMDB Scores by Content Type**

✅ **Query:**

```
SELECT type,
ROUND(AVG(imdb_score),2) AS avg_imdb_score,
ROUND(AVG(tmdb_score),2) AS avg_tmdb_score
FROM titles
GROUP BY type;
```

type	avg_imdb_score	avg_tmdb_score
SHOW	6.64	6.69
MOVIE	6.82	6.61



📊 **Explanation:**
This query calculates the average IMDB and TMDB scores for each content type (Movies and TV Shows). The ROUND() function is used to limit the values to 2 decimal places for better readability.
💡 **Insight:**
It helps compare the overall ratings of movies and shows, showing which type of content generally receives higher audience ratings.

🎯 **Step 4: Count of Movies and Shows by Decade**
✅ **Query:**

```
SELECT
   CONCAT(FLOOR(release_year / 10) * 10, 's') AS decade,
   COUNT(*) AS movies_shows_count
FROM titles
WHERE release_year >= 1940
GROUP BY CONCAT(FLOOR(release_year / 10) * 10, 's')
ORDER BY decade;
```

decade	movies_shows_count
1940s	1
1950s	5
1960s	8
1970s	18
1980s	41
1990s	6

📊 **Explanation:**
This query groups movies and shows into decades (like 1980s, 1990s) using the FLOOR() function. It then counts how many titles were released in each decade. The WHERE clause filters data from 1940 onwards.
💡 **Insight:**
It helps identify trends in content production over time, showing which decades had the highest number of releases.

🎯 **Step 5: Average IMDB and TMDB Scores by Age Certification**

✅ **Query:**

```
SELECT age_certification,
ROUND(AVG(imdb_score),2) AS avg_imdb_score,
ROUND(AVG(tmdb_score),2) AS avg_tmdb_score
FROM titles
WHERE age_certification IS NOT NULL
AND age_certification <> ''
GROUP BY age_certification
ORDER BY avg_imdb_score DESC;
```

age_certification	avg_imdb_score	avg_tmdb_score
TV-14	8.8	8.31
PG	7.67	7.44
PG-13	7.5	7.35
TV-PG	7.43	7.43
R	7.02	7.28
TV-G	6.9	7.4
TV-Y	6.67	7.07
TV-MA	4.05	3.5

📊 **Explanation:**
This query calculates the average IMDB and TMDB ratings for each age certification category (such as PG, R, etc.). The data is grouped by age certification and sorted by IMDB score in descending order.
💡 **Insight:**
It helps understand how content ratings relate to audience scores, showing which age categories tend to receive higher ratings.

🎯 **Step 6: Average Runtime by Content Type**

✅ **Query:**

```
SELECT
   type AS content_type,
   ROUND(AVG(runtime),2) AS avg_runtime_min
FROM titles
GROUP BY type;
```

content_type	avg_runtime_min
SHOW	27.82
MOVIE	115.28

📊 **Explanation:**
This query calculates the average runtime for each content type (Movies and Shows) by grouping the data based on the type column. The ROUND() function formats the result to 2 decimal places.
💡 **Insight:**
It provides a direct comparison of the average duration of movies and TV shows, helping to understand differences in content length.

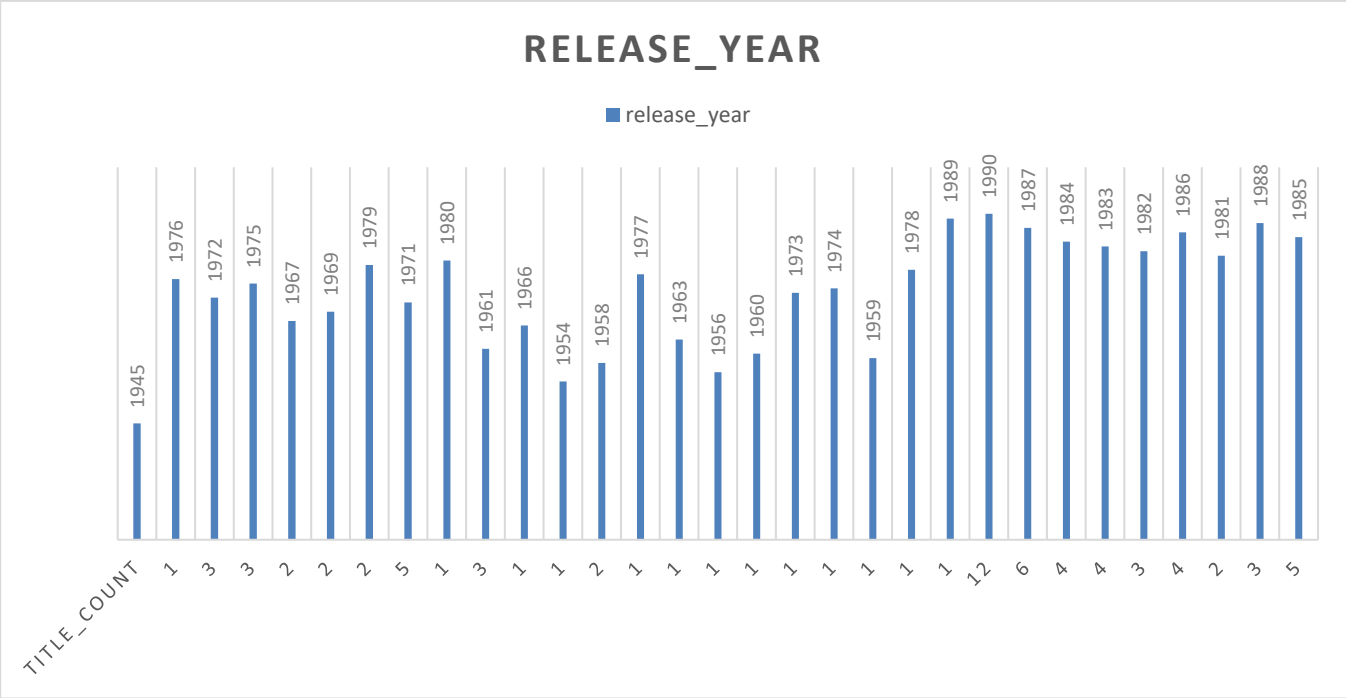
🎯 **Step 7: Top 10 Shows with the Most Seasons on Netflix**

✅ **Query:**

```
SELECT
   title,
   SUM(seasons) AS total_seasons
FROM titles
WHERE type = 'Show'
GROUP BY title
ORDER BY total_seasons DESC
LIMIT 10;
```

title	total_seasons
Wheel of Fortune	39
Thomas & Friends	24
Fireman Sam	13
Danger Mouse	10
Seinfeld	9
Saved by the Bell	5
Monty Python's Flying Circus	4
Knight Rider	4
Major Dad	4
Five Came Back: The Reference Films	1

📊 **Explanation:**
This query filters only TV shows and calculates the total number of seasons for each title. The results are sorted in descending order to identify the shows with the highest number of seasons, and only the top 10 are displayed.
💡 **Insight:**
It helps identify long-running shows on Netflix, which are often more popular or successful over time.



🎯 **Step 8: Total Number of Titles Released Each Year**

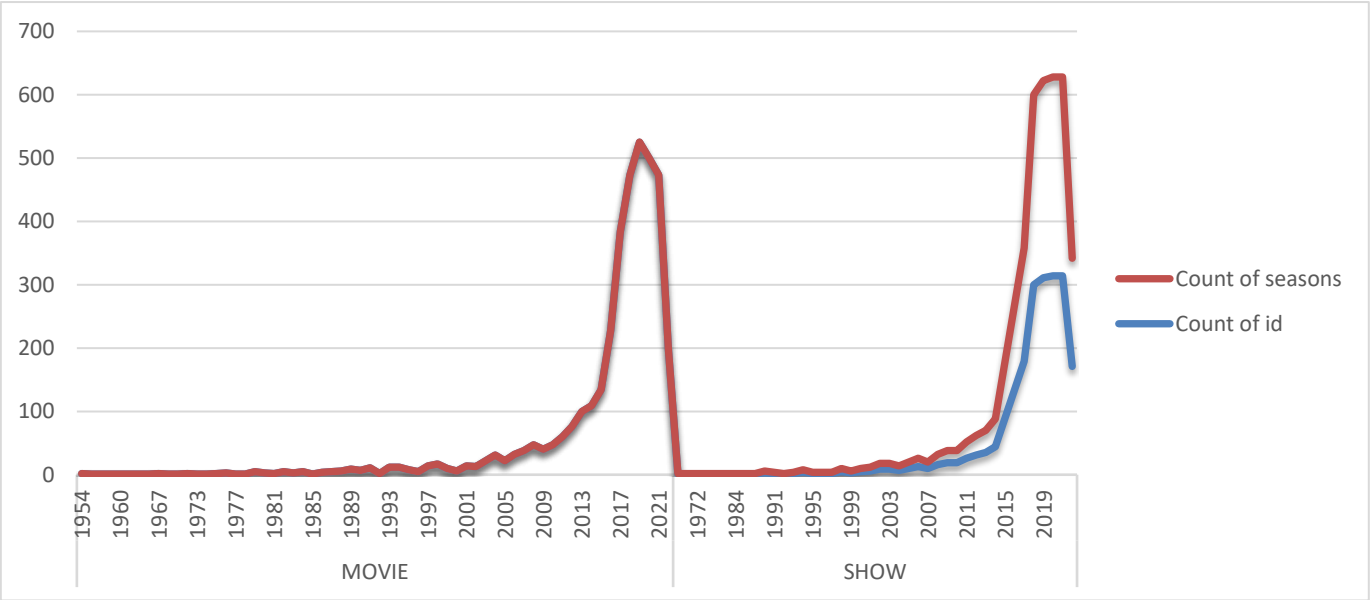
✅ **Query:**

```
SELECT
   release_year,
   COUNT(*) AS title_count
FROM titles
GROUP BY release_year
ORDER BY release_year DESC;
```

release_year	title_count
1990	6
1989	12
1988	5
1987	4
1986	2
1985	1
1984	4
1983	3
1982	4
1981	3
1980	3
1979	5
1978	1
1977	1

📊 **Explanation:**
This query counts how many titles were released in each year by grouping the dataset based on release_year. The results are sorted in descending order to show the most recent years first.
💡 **Insight:**

It helps identify trends in content production over time, such as growth in recent years or periods with fewer releases.



1 Key Insights

- Movies and TV shows differ in both average ratings and runtime
- Content production has significantly increased in recent decades
- Age certification influences audience ratings
- A small number of shows have very high season counts, indicating long-term popularity
- Recent years show a higher volume of content compared to earlier periods

2 Limitations of the Dataset

- Some columns contain NULL or missing values
- production_countries may include multiple countries in one row
- Data may not represent all global content accurately

3 Tools & Technologies Used

- SQL (MySQL)
- Dataset for movie analysis
- Query-based data exploration

4 Conclusion (FINAL SECTION)

This project demonstrates how SQL can be used to explore and analyze a dataset effectively. By performing data cleaning, aggregation, and trend analysis, meaningful insights were extracted from the movie dataset.

The analysis highlights patterns in content production, ratings, and distribution, showcasing practical skills required for a data analyst role.

This project demonstrates practical SQL skills such as data cleaning, aggregation, and analytical querying, which are essential for data analyst roles.